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Editors' Note

The *JOURNAL OF ANCIENT CIVILIZATIONS (JAC)* is published annually in two fascicles by the Institute for the History of Ancient Civilizations (IHAC, Northeast Normal University, Changchun, Jilin Province, People's Republic of China).

The aim of *JAC* is to provide a forum for the discussion of various aspects of the cultural and historical processes in the Ancient Near Eastern and Mediterranean world, encompassing studies of individual civilizations as well as common elements, contacts, and interactions among them (e.g., in such traditional fields as Assyriology, Egyptology, Hittitology, Classics, Byzantine Studies, and Sinology, among others). Hence, we publish the work of international scholars while also providing a showcase for the finest Chinese scholarship, and therefore welcome articles dealing with history, philology, art, archaeology, and linguistics that are intended to illuminate the material cultures and societies of the Ancient Near East, the Mediterranean region, and ancient China. Articles discussing other cultures will be considered for publication only if they are clearly relevant to the ancient Mediterranean world, the Near East, and China. Information about new discoveries and current scholarly events is also welcome. Publishers are encouraged to send review copies of books in the relevant fields.

JAC is a double-blind peer-reviewed journal. There is no APC charged. Articles must not have been published in, or submitted to, another publication at the time of submission. All articles submitted are first carefully read by at least two editors of *JAC*, who will provide feedback to the author. Articles (excluding book reviews or research reports) are subsequently reviewed anonymously by at least two referees in the specific field, appointed by the editorial board. The whole peer-review process as well as any judgment by peer-reviewers or editors is entirely based on the quality of the article and the research conducted therein. In cases where the reviewers or editors recommend changes in the manuscript, authors are requested to revise their articles. The final approval of articles is at the editorial board. Throughout the whole peer-review process, articles are treated confidentially. If significant errors or inaccuracies are detected during or after publication, it is any party's and in particular the author's obligation to promptly notify the journal editors and cooperate with the editors to correct or retract the paper. In case of (alleged or supposed) interest conflict, malpractice, misconduct, contested authorship, or plagiarism of any party involved in the double-blind peer-review process the editor-in-chief and/or the executive editor-

in-chief (or, if necessary, another member of the editorial board) will pursue the case and should the situation of taking action arise, will notify and hear the respective party before final decision by the editorial board. All articles not in accordance with the aforementioned standards will be removed from publication and indicated as such. From time to time, we will publish a list of the referees on our homepage (<http://ihac.nenu.edu.cn/JAC.htm>) to make the double-blind peer-review process transparent and comprehensible.

This fascicle features selected papers from two workshops held at IHAC, on “Future Developments of Digital Humanities and Didactics in Global Ancient Studies” (23–24 September 2024) and on “Virtual Reality and the Ancient World” (24–25 March 2025), detailed reports of which have been published elsewhere (Günther, S. 2024. “Conference Report: Future Developments of Digital Humanities and Didactics in Global Ancient Studies.” *H-Soz-Kult*, 08.11.2024, URL: <https://www.hsozkult.de/conferencereport/id/fdkn-151035>; id., “Virtual Reality and the Ancient World.” *H-Soz-Kult*, 05.05.2025, URL: <https://www.hsozkult.de/conferencereport/id/fdkn-154713> [07.10.2025]). In both meetings, the speakers and audience critically engaged with current digital trends (not only) in ancient studies, namely, to access, study, and teach the ancient world via digital tools, with virtual reality setups, and through artificial intelligence tools. With the publication, we hope to promote the discussion about the chances and challenges of digital methods in the didactics of ancient studies.

All communications, manuscripts, disks, and books for review should be sent to the Assistant Editor, Journal of Ancient Civilizations, Institute for the History of Ancient Civilizations, Northeast Normal University, 130024 Changchun, Jilin Province, People's Republic of China (e-mail: jac@nenu.edu.cn), or to the Executive Editor-in-Chief, Prof. Dr. Sven Günther, M.A. (e-mail: svenguenter@nenu.edu.cn or sveneca@aol.com).

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TEACHING AI IN ARCHAEOLOGY:
PRACTISING PATTERN RECOGNITION WITH ORANGE DATA MINING

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Introduction

AI has become extremely popular within the past few years and is predicted to be both the saviour and the downfall of mankind.¹ In fact, it is already an inescapable part of our daily lives, running (more or less openly) in the background of many applications which are used on a regularly basis: text and image generators, driving assistants, photo filters in smartphones, co-pilots for writing, Google images, AI stocks, and a number of chatbots, to name just a few biotopes of AI.

AI has also entered the field of Classics and many other traditional research fields. Among the most recent publicly available applications the most notable for Classics are: Ithaca (restoring Greek inscriptions),² Aeneas (restoring Latin inscriptions),³ Humanitext Antiqua (analysing and interpreting ancient texts),⁴ ArchAIDE (classifying potsherds),⁵ and xCurator (creating virtual exhibitions

¹ See, e.g., URL: <https://www.safe.ai/work/statement-on-ai-risk> (04.04.2025); Jiang et al. 2022; Hendrycks, Mazeika and Woodside 2023.

² Collaborators: google DeepMind, Università Ca'Foscari Venezia, University of Oxford, Athens University of Economics and Business; see URL: <https://ithaca.deepmind.com> (04.04.2025); Assael et al. 2022. The dataset used is the one of the Packard Humanities Institute's epigraphic database (PHI), see URL: <https://inscriptions.packhum.org> (04.04.2025).

³ Collaborators: google DeepMind, Durham University, University of Nottingham, University of Warwick, University of Oxford, Athens University of Economics and Business, google cloud, Sint-Lievenscollege; see URL: <https://predictingthepast.com/aeneas> (20.08.2025); Assael et al. 2025. The dataset comprises of several large epigraphic databases, see URL: <https://github.com/google-deepmind/predictingthepast> (20.08.2025) and Assael et al. 2025.

⁴ Collaborators: Naoya Iwata (Nagoya University); Ikko Tanaka (J. F. Oberlin University); Jun Ogawa (National Institute of Informatics); see URL: <https://humanitext.ai> (04.04.2025); Iwata, Tanaka and Ogawa 2024; the dataset used is the one of the Perseus Digital Library, see URL: <https://www.perseus.tufts.edu/hopper> (04.04.2025).

⁵ The ArchAIDE Project has received funding from the European Union's Horizon 2020 research and innovation programme. Collaborators: Mappa Lab of the Università di Pisa, Visual Computing Lab of the Consiglio Nazionale delle Ricerche, Deep Learning Lab of the University of Tel Aviv, Archaeological Data Service of the University of New York, ARQUB Material Culture and Archaeometry of the Universitat de Barcelona, and CoDArchLab – Archaeological Institute of the Universität

with the help of AI-created object tags).⁶ A number of research projects dealing with the potential of AI for the study of ancient imagery and texts are critically reflecting its limits, in particular with regard to conventional classification methods.⁷ Moreover, the use of generative AI chatbots – especially ChatGPT – has considerably increased. Students (and, presumably, also scholars) understand them as relatively reliable tools for searching information and literature, for translating texts, and as a source of inspiration for writing attractive and orthographically correct texts. This goes hand in hand with the fear of AI cheating in written essays and the growing discomfort of academics regarding the question of how to deal with AI tools, especially generative AI, in class.

Despite the increasing⁸ role of AI in both academic teaching and research it is, at least in Humanities, rarely a topic openly discussed in class. With regard to Classics, it is, as far as I know, very much underrepresented in courses.⁹ As such, I decided to change this by teaching two courses in the winter semester 2024/25 at the University of Heidelberg, which focus on the potential and limitations of AI applications for Classics and Archaeology (“KI in den Altertumswissenschaften und der Klassischen Archäologie”).¹⁰ Since my research focuses on ancient imagery, I chose to focus on pattern recognition of Greek and Roman imagery and objects to confine the complex content (i.e., excluding, for example, the use of AI for site recognition and in excavations). During the course for advanced students (“Hauptseminar”), we engaged in methodological

zu Köln; see URL: <https://www.archaide.eu>; <https://archaide-desktop.inera.it> (04.04.2025); Anichini 2021.

⁶ Collaborators: Badisches Landesmuseum Karlsruhe, Allard Pierson Museum Amsterdam; see URL: <https://xcurator.landesmuseum.de> (04.04.2025); Thiel and Posthumus 2023.

⁷ Exemplary with regard to methodological reflections are three German AI projects in the field of pattern recognition: ClaReNet (classifying Celtic coins), see URL: <https://clarenet.hypotheses.org> (04.04.2025); Egraphsen (attributing Greek vase paintings), see URL: <https://www.uni-goettingen.de/de/598165.html> (04.04.2025); Iconographics (iconographic analysis of vase paintings with pose estimation), see URL: <https://www.klassischearchaeologie.phil.fau.de/projekt-iconographics> (04.04.2025). For the current debate in scholarship, see also Günther et al. 2023a.

⁸ For the “post-digital” age and the role of AI in archaeology, see Gattiglia 2025, while for some critical notes regarding the use of AI for historical studies, see Kansteiner 2022. For the debate among scholars about the potential and limits of teaching digital archaeology see, e.g., Günther et al. 2023b.

⁹ Although there are some courses available, e.g., the Digital Classicists Seminars by the Berlin-Brandenburgische Akademie der Wissenschaften, with focus on AI, see URL: <https://www.hsozkult.de/event/id/event-139472> (04.04.2025). For the didactic potential of AI with regard to translating ancient texts, see Díaz-Sánchez and Chapinal-Heras 2024. Very recently, a new journal focusing on the teaching of AI applications in Classics, *AI & Antiquity*, has been launched by the Center for Innovation in Ancient Worlds, see URL: <https://ai-antiquity.org/index.php/ai> (04.10.2025).

¹⁰ URL: <https://www.uni-heidelberg.de/de/forschung/forschungsprofil/fields-of-focus/field-of-focus-iii/forschungsschwerpunkte/heidelberg-center-for-digital-humanities-hcdh/weiterbildung-digital-humanities/semesterprogramm-cas-hcdh-wise-20242025> (04.04.2025).

and ethical¹¹ questions and dealt with current research projects employing AI. The exercise course (“Übung”) served to practically explore, evaluate and discuss available AI tools (see the list above), and to practise pattern recognition using Orange Data Mining¹² which offers simple but powerful widgets that allow individuals to cluster and classify images and build a workflow to identify unknown images based on the user’s dataset.

In this paper, I will outline the use of Orange Data Mining for pattern recognition on ancient imagery and demonstrate how it can serve as a useful tool in class to facilitate the discussion of AI applications and limitations in archaeological research.

Pattern recognition with Orange Data Mining

Orange Data Mining (currently Orange 3) is an open-source program which was developed a decade ago by the Faculty of Computer and Information Science at the University of Ljubljana for statistical analysis, especially for the field of Bioinformatics.¹³ Thanks to number of add-ons (“widgets”), Orange 3 is adjustable to nearly all kinds of (big) data analysis and machine learning.

For a classroom full of students (and a teacher) of Humanities, it comes in handy that no coding skills are required. Orange 3 is based on “visual programming,” which means that the user can arrange a number of stacking components (widgets) and connect them to build a workflow (analytic pipeline) according to their own needs.¹⁴ The developers offer easily accessible explanations of the widgets (widget catalogue)¹⁵ and even a number of YouTube videos for beginners.¹⁶ The add-on Image Analytics is made for pattern recognition with a choice of different algorithms (embedders).¹⁷ However, before explaining this, let us step back a little to explain what pattern recognition actually is and how it works.

Pattern recognition is a field of machine learning which is devoted to the

¹¹ Ethical discussion included the questions of copyright (see, e.g., Crawford and Paglen 2021, for a critical examination of the image net database with is used by Orange Datamining for pretraining), the danger of “racist” AI (see, e.g., Hofmann et al. 2024), and the impact of AI on the environment (see United Nations Environment Programme 2024). These issues were also addressed in the “Übung,” but less intensively due to the German concept of this teaching mode which focuses on practical hands-on training.

¹² Version 3.38. On Orange Datamining, see Demsar 2013.

¹³ URL: <https://orangedatamining.com>; <https://github.com/biolab/orange3>; <https://www.fri.uni-lj.si/en/laboratory/biolab-27> (04.04.2025).

¹⁴ URL: <https://orangedatamining.com/home/visual-programming> (04.04.2025).

¹⁵ URL: <https://orangedatamining.com/widget-catalog> (04.04.2025).

¹⁶ URL: <https://www.youtube.com/channel/UCIKKWBe2SCAEyv7ZNGHle4g> (04.04.2025).

¹⁷ URL: <https://orangedatamining.com/widget-catalog/text-mining/corpus-widget> (04.04.2025).

identification and classification of patterns which appear in images or texts. To this end, artificial neural networks are trained on a big dataset of images (or texts), and the overall aim is to classify figures or objects shown in the imagery (or words/phrases in a text). This means that, turning to image analysis, a number of classes have to be defined beforehand, e.g., archaeological vessels such as “amphora,” “jug,” and “krater,” and the algorithm will then sort the respective images into these pre-defined classes; more precisely, the algorithm will assess the probability of an image belonging to one of the pre-defined classes. For example, a vessel shown in an image belongs to the class “amphora” – this is the so-called “ground truth.” If the algorithm predicts that the image belongs to this class by, e.g. 97 %, it is correct. Correct and wrong predictions – “correct” and “wrong” being based on the ground truth – can be used to provide feedback within the model which then adjusts the “weight” of specific patterns for the computing of later output rounds.

Fig. 1 shows a visualisation of this complex process in a simplified manner. An image of an amphora might serve as input data, more precisely the single pixels (in case of RGB photos these pixels come in three layers, that is, red, green and blue). In the first layers of the algorithm (the very first is called “input layer”), the computer recognises patterns based on the lines and edges in the image. Mathematically, this is done by matrices which “filter” out the small parts of the image which contain such patterns and generate “feature maps” (each “map” highlights the presence and location of a specific feature). While the algorithm repeats the recognition of patterns during the next layers, it also decreases the size of the image (“pooling”), until “flat” output vectors emerge which predict the attribution of the input image to one of the defined classes. The layers are “folded” for interconnection, and this is why this kind of neural network is called Convolutional Neural Network (CNNs).¹⁸ The number of layers depends on the respective “architecture” of the layers: for instance, VGG 16, a network commonly used for image pattern recognition, has sixteen layers.¹⁹ Like in a biological neural network, not all information will pass through, and not each and every neuron is connected with each other. Depending on the filtering matrices, only relevant features – relevant for the “correct” identification of the image based on the ground truth – will be passed on. Then the algorithm “learns,” based on its correct classifications – via feedback loops – which features are relevant, and which are not, and adjusts itself according to this training.

¹⁸ For a comprehensive introduction, see Bishop and Bishop 2024; Bengio 2009; for an easily accessible overview, see the Google Machine Learning Education, URL: <https://developers.google.com/machine-learning/practica/image-classification/convolutional-neural-networks> (04.04.2025).

¹⁹ URL: <https://github.com/ashushekar/VGG16> (07.05.2025); Simonyan and Zisserman 2015.

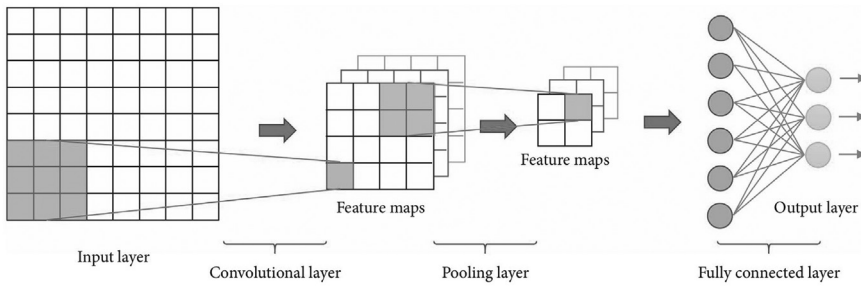


Fig. 1: Pattern recognition with an artificial neural network (from Meruane et al. 2021, fig. 1).

This means that usually a large training set is needed for achieving sufficiently accurate results. For the best results, the training set should be made up of between ten to a hundred thousand images. Turning back to Orange Data Mining, the extensive training set required causes an issue due to a user intending to analyse a certain number of images but being unable to train a CNN on such a large scale. In addition, deep (i.e., containing many layers) neural networks require a computing performance which would exceed the limits of an ordinary personal computer. As such, the developers offer an “embedding” of images. This works in the following way. The respective model – Orange 3, offering seven different neural networks to choose from²⁰ – has been pre-trained on a specific dataset, e.g., VGG 19 with the dataset “image net” (more than 14 million images)²¹ which consists of all kinds of images. Only the penultimate layer of the model, which is already a “flat” vector as explained above, is implemented in the widget as an embedder, and can be trained on the user’s much smaller dataset.²² This “small-scale” machine learning works surprisingly well, as we shall see below. However, and this is very relevant for someone dealing with archaeological material, the data set used for the pre-training cannot be accessed or even modified by the Orange 3 user, and this image training has, of course, not been done with ancient imagery. In addition, the architecture and the “weights” controlling the adjustment in the learning process are also not specified for archaeological needs. This affects the results of the image analysis widgets, as we will see soon.

²⁰ URL: <https://orangedatamining.com/widget-catalog/image-analytics/imageembedding> (07.04.2025).

²¹ URL: <https://image-net.org> (07.04.2025).

²² Godec et al. 2019.

“Image Grids” and the “black box” phenomenon

The first widget we explored in class was “Image Grids.” The analytic pipeline has to be arranged in the following way (cf. Fig. 2). With “Import Images,” the user can import desired images to Orange 3. These images are then arranged into folders, the name of each folder defining the class to constitute the ground truth. With “Image Viewer,” the user may take a look at the images. If one links the widget “Import Images” with “Image Embedding,” the image data will be embedded. The user may then choose between seven different embedders (see the drop-down menu in Fig. 2). The table in the lower right shows how the embedder works: for each image (file name under “image”) it computes a row of features (n0, n1, n2, ..., n1000), depending on the respective embedder chosen. When linking “Image Embedding” with the widget “Image Grid,” Orange 3 will order the images in a grid, depending on their features.



Fig. 2: Widget arrangement to produce an image grid with Orange 3. Screenshot by the author.

Fig. 3 displays the image grids based on features which have been generated by the embedders DeepLoc and VGG 19. Images that are considered to be more similar are placed closer to each other, those less similar are shown with more distance between them. This means that the students could try to figure out which features are more relevant for which embedder, and this is what we discussed in class based on the students’ results. The 43 images used for this task were 42

photos and one drawing of Attic cups seen from above, mostly focusing on the image of the inner part (tondo), taken from the Beazley Archive Pottery Database under the term “satyr” with regard to the vessels’ decoration.²³ The premise for this small dataset was to provide images which show similar motifs, but are of a certain variety. Mostly one single satyr, sometimes two satyrs, and in four cases a gorgoneion (which means that satyrs were depicted on the outside of the cup, not inside) were painted on the cup’s inner part. Both red and black figure were included in the dataset, and some cups were fragmented while some photos showed the full cup including the handles.

The differences between the two grids are quite obvious (Fig. 3). The grid with DeepLoc as the embedder consists of mainly two clusters, one left, one right, and a few images singled out. At first glance it is obvious that the brightness/darkness of the single images seems to be a key feature for sorting by DeepLoc. In addition, the dispersion of dark and bright segments plays an important role: the left cluster consists of images whose frames are “filled” with black, while in the right cluster, a dark circle-like object (either tondo, full cup or fragment) is enclosed in a bright background. The few images singled out deviate from this pattern: the drawing on the upper right does not include any black segments, the black figure tondo between the clusters is a greyish circle enclosed in black, and the only colour photo below shows a black circle “cut off” by the frame on three sides. Since DeepLoc was trained to identify the localisation of proteins in yeast cells – more precisely, the location and abundance of fluorescently

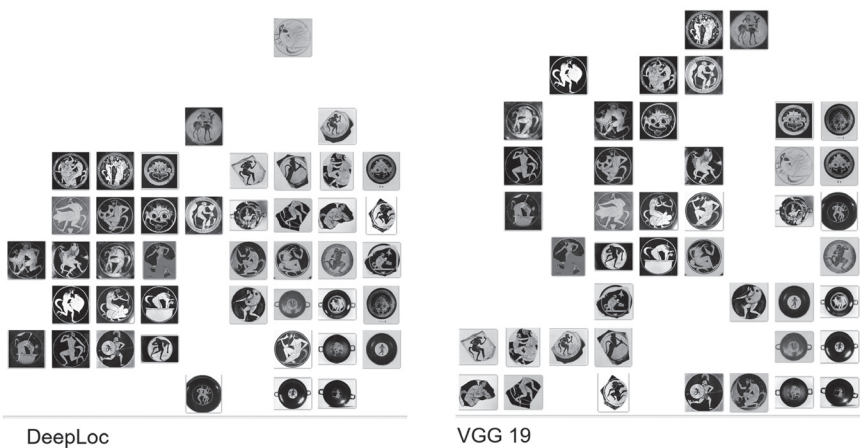


Fig. 3: Image grids with the insides of 43 Attic drinking cups using two different embedders, DeepLoc (left) and VGG 19 (right). Screenshot by the author.

²³ URL: <https://www.carc.ox.ac.uk/carc/pottery> (07.04.2025). The “decoration termword” was “satyr.”

labelled proteins²⁴ – black-and-white contrast, simple shapes and especially the spatial location of both (for the purpose of detecting subcellular location) play an important role for DeepLoc.

The grid based on VGG 19 is more complex than the one done with DeepLoc. However, one can nevertheless evaluate some features that seem to have been more relevant for the algorithm. The black-and-white contrast again seems to bear some weight in pattern recognition, but not as much as it did for DeepLoc. In addition, the outline seems to play a significant part since all the fragments are grouped together (lower left) as well as all cups captured in total (lower right). Furthermore, three of the four gorgons – human-like faces – are placed next to each other (upper right). The complexity of the image in total seems also to have been taken into account since the tondos with more than one figure and a more refined meander are shown in proximity (upper middle). Again, this does not come as a surprise since VGG 19 was made for image analysis, in which the shape of objects, of human faces and the complexity of patterns are expected to play an important role. Both for VGG 19 and for DeepLoc, colour information does not appear to be relevant since the only colour photo was not clearly separated from the others. This too is not surprising as categorising colours is an irrelevant feature for both the pattern recognition of humans, animals and objects in a photo, and for the localisation of fluorescent proteins within a yeast cell.

In summary, in this first round, the students learned that final results clearly depended on the structure of the chosen embedder. They evaluated and discussed the importance of specific features for the grid output, and reflected on the limits of the models' explainability, because, in the very end, one cannot be sure about the weight of specific features learned by the algorithm. In this way, the students experienced the "black box" phenomenon,²⁵ and this was important to develop a critical awareness regarding the employment of AI for archaeological research in class. In a further, more methodological, step the biases caused by the choice of the dataset – for example, the ethical and political issues inscribed into the "image net" dataset – were discussed by the students of the seminar class.²⁶

²⁴ Almagro Armenteros et al. 2017; Kraus et al. 2017.

²⁵ An approach to allow insights into the black box is Grad-CAM (Gradient-weighted Class Activation Mapping), localising the important regions in the image and visualising them with heatmaps: Selvaraju et al. 2020.

²⁶ For a critical evaluation of the image net database, see Crawford and Paglen 2021.

The importance of the dataset: “Image Clustering”

The second analytic pipeline introduced to the students was “Image Clustering.” For this task, the widgets “Distances”²⁷ and “Hierarchical Clustering”²⁸ (both belonging to the “Unsupervised” widget group rather than “Image Analysis,” which can be employed for different kinds of data analysis) were connected to “Image Embedding,” and produced a dendrogram of images, depending on their visual similarity (distances of the features generated with the respective embedder). Again, variations between the different embedders were discussed by the students, and the “black box” phenomenon played a role here as well.

The focus of this exercise was, however, to draw the students’ attention to the importance of the dataset. Instead of cup tondos, we used images of 69 Greek vases with different shapes, defining the respective vase shapes to be our classes: amphora, guttus, jug, krater, kylix (cup) and lekythos. The source of the images was again the Beazley Archive Pottery Database. As Fig. 4 demonstrates, this worked relatively well; made obvious in the cut-out of the (much bigger) dendrogram in the upper part. Images whose branches divide more to the right are considered to be more similar, and therefore have less distance. Branch C1 (with highlighted background) groups one lekythos, seven cups (kylikes) and eleven kraters. And yet, images depicting the very same vase shape also appear in branches with higher distance (e.g., the group of kylikes (cups) in the branches above), which is – in terms of vase shape clustering – wrong. However, within branch C1, cups and kraters are sorted “correctly” into two groups, branching out quite early (second branching), and the only lekythos in C1 stems into a single branch with higher distance – which is correct. One may also note that the distances between the single kraters is higher than between the single cups, and this in keeping with the fact that the class “krater” indeed has more variety, since it both includes bell kraters and calyx kraters.

Nonetheless, one wonders why a lekythos is part of branch C1, and not placed in one branch together with the other lekythoi. A comparison with the group of vases in Fig. 4 to the left (three kraters, four cups and the lekythos of branch C1) helps us uncover explanations for this “mistake:” (1) it is obvious that the lekythos is only preserved in its lower part, neck and rim are missing (compare the two fully preserved lekythoi in Fig. 4 to the right), thus the shape of the C1 lekythos appears to be less slim, and the outline deviates from the fully preserved lekythoi; (2) the resolution of the photo showing the C1 lekythos is very low, which might have blurred the features; (3) the photo of the C1 lekythos was

²⁷ URL: <https://orangedatamining.com/widget-catalog/unsupervised/distances> (07.04.2025).

²⁸ URL: <https://orangedatamining.com/widget-catalog/unsupervised/hierarchicalclustering> (07.04.2025).

taken from a different perspective than the two fully preserved lekythoi, and thus the shape of the foot more closely resembles the foot of one of the kraters and/or cups.

There are, however, also some queries and quandaries which remain and cannot be solved due to the algorithm’s “black box:” (1) although the C1 lekythos appears to be less slim than the other lekythoi, it is still not as flat as the cups which are closest in the dendrogram, even closer than the kraters; (2) not all photos of lekythoi in the full dataset were taken from above, but also some in frontal view, thus the foot of the C1 lekythos should not differ that much from the other lekythoi; (3) one wonders whether the decoration of the vessels was considered by the algorithm since in this regard the lekythos resembles rather the complex figures on many cups than the (mostly) more simply painted kraters.



Fig. 4: Orange 3 dendrogram of Greek vase shapes with branch C1 highlighted, and below photos of vases sorted into C1 (group on the left), and two fully preserved lekythoi (group on the right). Screenshot by the author.

Based on the results of this second data pipeline, the students again discussed the “black box” phenomenon and the intransparency of neural networks. In addition, they were sensitised to the impact of the user’s dataset which was the basis for image clustering. Presumably, the preservation state of the archaeological object, the angle of the photo camera and the quality of the images – very often the photos available are regrettably blurred and of low resolution – affected

the dendrogram. The availability of high-quality photos in a sufficient number and from the same viewpoint is, indeed, another significant limitation for the application of pattern recognition in current archaeological research.²⁹ This was further discussed in the seminar class, focusing on current pattern recognition publications in the field of Classics and Archaeology.³⁰

Scoring the model: “Image Classification”

The next data training (Fig. 5) was devoted to “Image Classification,” again using the small dataset of Greek vessels and the classes defined for the aforementioned “Image Classification” round (amphora, guttus, jug, krater, kylix/cup, and lekythos). This time, however, the images were not arranged according to their distances but had to be sorted into the pre-defined classes. The classification was done by two widgets (widget group: “Model”), “Logistic Regression”³¹ and “Neural Network,”³² based on “Image Embedding.” While “Logistic Regression” is a commonly used statistical model which serves to classify data,³³ “Neural Network” employs a multi-layer perceptron algorithm with backpropagation, and the possibility of adjusting certain parameters of the network.

The data processed by “Image Embedding” and the two model widgets were then connected to the widget “Test and Score,”³⁴ which enables the user to compare the performance of both “Logistic Regression” and “Neural Network.” The results were then presented in a confusion matrix by the respectively named widget,³⁵ and “Image Viewer” helped to display images with incorrect prediction.

The “Confusion Matrix” (Fig. 5 on the upper right) provides an overview of the classification result (in this case done by “Logistic Regression”). 68 out of 69 images were classified correctly (diagonal line with highlighted background). Only one vessel was predicted incorrectly, claiming it to be a guttus, but according to the ground truth it was actually a cup. This surprisingly good result is reflected in the evaluation result of the widget “Test and Score.” The AUC score,³⁶ a proxy for the model’s performance, is 0.999 (a score of 1 meaning that

²⁹ See, e.g., the discussion in Günther et al. 2023a.

³⁰ See, e.g., the discussion of the challenges regarding the dataset for the Archaide project: Anichini et al. 2021; for the organisation of the dataset and LOD (Linked Open Data), see URL: <https://archaide-desktop.inera.it/blog/-/blogs/linking-it-all-together?groupId=20126&pk=2482318&userId=228845> (07.04.2025).

³¹ URL: <https://orangedatamining.com/widget-catalog/model/logisticregression> (07.04.2025).

³² URL: <https://orangedatamining.com/widget-catalog/model/neuralnetwork> (07.04.2025).

³³ URL: https://en.wikipedia.org/wiki/Logistic_regression (07.04.2025).

³⁴ URL: <https://orangedatamining.com/widget-catalog/evaluate/testandscore> (07.04.2025).

³⁵ URL: <https://orangedatamining.com/widget-catalog/evaluate/confusionmatrix> (07.04.2025).

³⁶ A very comprehensive explanation is provided via URL: <https://developers.google.com/machine-learning/crash-course/classification/roc-and-auc> (07.04.2025).

all predictions are correct). The neural network’s performance was a bit lower, 0.994, demonstrating that neural networks are not the better choice in all cases. It stands to reason that, depending on the courses’ time frame, students could manipulate the parameters of both the logistic regression model and the neural network in an attempt to increase their respective performance indicated by the AUC score. In our class, however, we chose a different approach.

For the purpose of our class, students started their own “experiments” in small teams of two or three, with the aim to classify ancient images and objects, and to raise the AUC score of the models as much as possible by experimenting with different sizes of their dataset, the choice of images in the dataset, and the number and size of the classes defined. These experiments provided some very interesting and elucidating results. Among the objects and images, freely chosen by the respective teams, were fibulas, sculptures, Roman coins, Roman portraits or drawings of Bronze age seals. The choice depended very much on the availability of a sufficient amount of photographs with high quality features and standardised perspectives (which is, as already mentioned, an important issue in archaeological AI research).

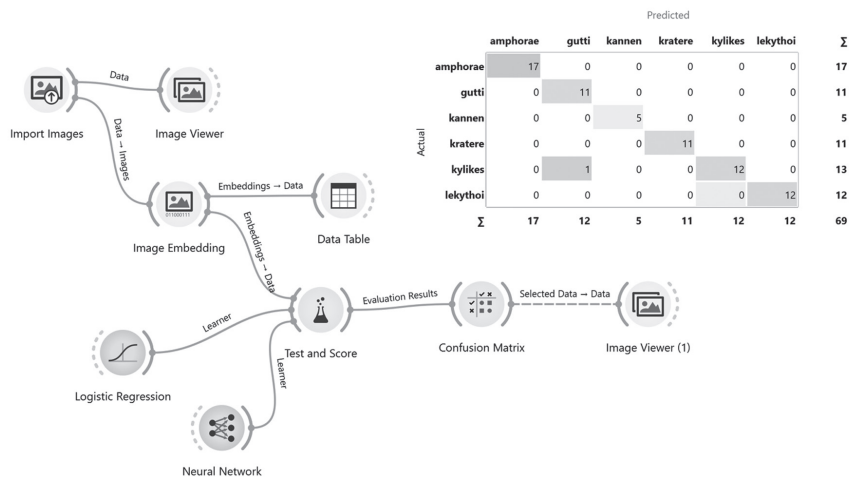


Fig. 5: Orange 3 “Image Classification” data pipeline (left) and “Confusion Matrix” (right). Screenshot by the author.

There were considerable variances in the results, depending on the material chosen, which I will quickly outline based on two experiments. The first experiment was done by two students with Roman imperial coins. Their aim was to identify the respective emperor shown on the obverse. Their first try with 50 coins and ten classes was quite poor (the AUC score of “Logistic Regression”

was 0.711; 0.5 equaling random distribution),³⁷ which is not surprising since every class contained only 5 coins. The students discovered that both a higher number of coins per class (10 instead of 5) and a lower number of classes (3 instead of 10) led to much better results (the AUC score of “Logistic Regression” was raised to 0.908). However, there were still a considerable number of mistakes. It is interesting that coins of the emperors Geta, Nero and Septimius Severus (the 3 classes defined by the students) were confused, as Septimius Severus has a very distinct, curly beard in his portrait, while Nero and Geta are both portrayed as young men with smooth cheeks (see the photos of coins in Fig. 6). While the students assumed that this feature would be of relevance for the algorithm and thus raise the AUC score, it had obviously no impact. This impressively demonstrated how much human vision (and archaeologically trained eyes) and machine vision can differ from each other.³⁸

The second experiment I would like to mention, also performed by two students, was devoted to the drawings of Greek Bronze age seals, based on the open-source dataset of the Heidelberg CMS (*Corpus of Minoan and Mycenaean Seals*).³⁹ They defined three classes – namely, seals depicting figures, objects, and geometric patterns – and selected 65 images. The AUC score of the “Logistic Regression” was 0.902, and the one of “Neural Network” was even higher in this case at 0.919.

All groups presented their experiments and evaluated them. In all cases, a low number of classes, a sufficient number of images per class and a balanced class-size⁴⁰ raised the performances of the models, regardless of the material chosen. Nevertheless (and this was an important issue in the discussion following the presentations) while image classification worked well with some materials, like the seal drawings, it was much less successful with others, like Roman coins. Due to the “black box” phenomenon, reason(s) could not be fully and satisfyingly explored. However, it seems that Roman coins are difficult to classify due to the enormous complexity of their imagery patterns. Furthermore, due to their irregular circular shape, the outline of the object is a less distinct (and thus less useful) feature for identification and classification as, for instance, in case of Greek vase shapes. The seal drawings, in contrast, consist of simple lines in a

³⁷ They used the images available via OCRE (*Online Coins of the Roman Empire*), URL: <https://numismatics.org/ocre> (07.04.2025).

³⁸ For the difference between machine learning and human visual perception, see Wichmann and Geirhos 2023.

³⁹ URL: <https://books.ub.uni-heidelberg.de/propylaeum/partner/cms> (07.04.2025). For the dataset, see URL: <https://arachne.dainst.org/search?q=CMS> (07.04.2025).

⁴⁰ For overfitting and underfitting in the training, see, e.g., URL: <https://developers.google.com/machine-learning/crash-course/overfitting> (07.04.2025).

round or ovoid frame, which are only slightly shaded (cf. Fig. 6). This reduced design with clear lines and edges presumably favours correct pattern recognition, both for the embedder and the respective model (“Logistic Regression” or “Neural Network”).

Amphora by 97%? Working with the “Prediction” widget

The final in class task was to build a data pipeline which can correctly predict the class of an unknown image (Fig. 6).⁴¹ We started again with our dataset of Greek vases. The upper part of the data pipeline was nearly identical to the one employed for “Image Classification” (Fig. 5), except for the addition of the widget “Select Columns”⁴² (to define the columns on which the predictions will be based, in our case, the image features as calculated by the embedder). The model used for classification was “Logistic Regression,” although one could have, of course, chosen “Neural Network” instead. In the lower part, “Import Images” uploads images with undefined classes, that is, the images that should be embedded, which are then embedded by “Image Embedder” (it must be the same embedder as in the upper pipeline, of course). Both pipelines are merged

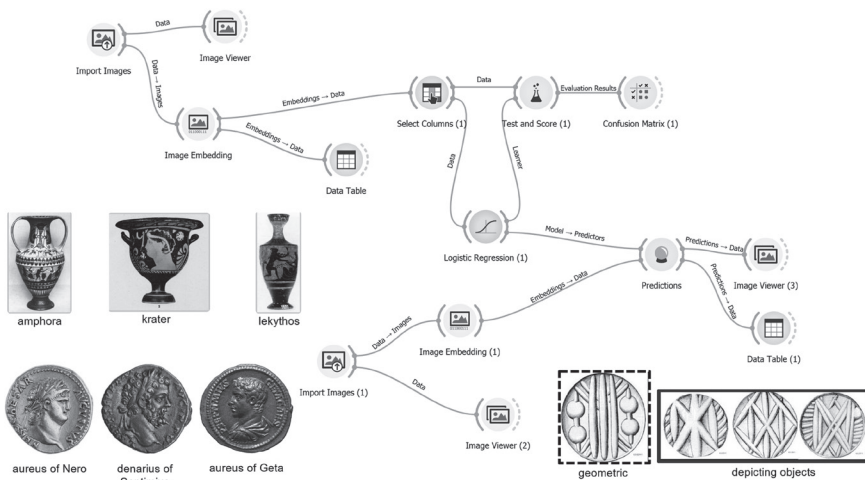


Fig. 6: “Image Prediction” data pipeline. Unknown images: Greek vases (upper left), Roman imperial coins (lower left) and drawings of Bronze age seals (right). Screenshot and pasticcio by the author.

⁴¹ It was based on the data pipeline developed in an Orange Data Mining workshop for archaeologists in 2021, see URL: <https://orangedatamining.com/blog/data-mining-for-archaeologists-part-i> (07.04.2025).

⁴² URL: <https://orangedatamining.com/widget-catalog/transform/selectcolumns> (07.04.2025).

with the widget “Predictions,”⁴³ which classifies the unknown image according to the upper pipeline. The results are made visible with a data table and “Image Viewer.”

The images with unknown classes are shown in Fig. 6 (on the left). All were correctly classified: according to “Predictions,” the first vessel (left) showed a 97 % probability of being an amphora, the second (middle) a krater with 95 % probability, and the third (right) a lekythos with 96 % probability. The students were then asked to reproduce the data pipeline with their dataset of the “Image Classification” experiment in order to test the degree to which the prediction worked with their respective material. Following which, the students were asked to evaluate any wrongly classified images and discuss the reason(s) why these mistakes might have occurred.

In the first experiment outlined above, which focused on Roman emperors on coins, three unknown coins were tested. A golden aureus of Nero, a silver denarius of Septimius Severus and a golden aureus of Geta were all predicted correctly. However, taking a look at the probabilities, the seemingly perfect result was shaken a bit. The first aureus (of Nero) was said to be Nero’s with only 71 % probability, the denarius showed only a 63 % probability of being Septimius Severus’, and the aureus of Geta was said to be Geta’s with 68 % probability, to be Nero’s with 21 % probability and to be Septimius Severus’ with 12 % probability. An interesting case was provided by the second experiment described above, namely, the one classifying Bronze age seal drawings. One of the images (lower right with dotted frame) was wrongly predicted to show an object, although it typologically belongs to the group of geometric ornaments. The comparison with three seal drawings that are said to be depictions of objects (stylised double axes), however, reveals a clear similarity in diagonal strokes. Thus, this seeming “mistake” demonstrates that the prediction worked very well from a structural, of course not human-archaeological, viewpoint.

Conclusion

Orange Data Mining proved to be a great teaching and learning in class tool, easily accessible for students (and teachers) without coding skills. Even within the strict time frame of one semester, students were able to explore pattern recognition and become more familiar with AI terminology and methodology. Not only did they develop the skills necessary to understand and discuss how AI works and how it challenges research in the Humanities, but they also critically reflected on the differences between machine learning and human visual

⁴³ URL: <https://orangedatamining.com/widget-catalog/evaluate/predictions> (07.04.2025).

perception.⁴⁴ Furthermore, they were able to perform their own experiments based on personal academic interests in archaeological material, and could practically explore the impact of different algorithms, datasets, and defined classes on their results.

By doing so, Orange Data Mining allowed the students to reproduce and reflect the challenges, limitations and potential of pattern recognition for archaeological research in a nutshell. With this knowledge, they were and hopefully will be further able to discuss and question conventional archaeological classifications, as well as new AI approaches in the (near?) future.

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⁴⁴ On the challenges in accessing ancient languages with machine learning, see, e.g., Sommershield 2023.

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ABSTRACTS

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CLASSICISTS IN THE DIGITAL AGE: TOWARD A *DIGITAL REPUBLIC OF LETTERS*? (pp. 117–139)

doi: 10.16758/j.cnki.1004-9371.2025.04.015

This paper discusses the ways in which classicists communicate and present themselves in the digital world by comparing selected “Western” and Chinese social media platforms and specific forms of digital communication. It argues that a *Digital Republic of Letters* is emerging, encompassing all the pros and cons of digitality alongside the very same participatory possibilities and limitations which the historical *Republic of Letters* offered. Therefore, it is natural to question whether this said *Digital Republic of Letters* will form a “Global Classics” community as it shows significant differences in the modes of how Greco-Roman Classics are assessed in different cultural frameworks.

Keywords: digital age – Global Classics – cultural frameworks – social media – *Digital Republic of Letters*

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THE PROJECT OF A CONNECTIVE INDO-EUROPEAN ETYMOLOGICAL ONLINE DICTIONARY: NEW PERSPECTIVES FOR ETYMOLOGICAL RESEARCH AND OUTREACH IN THE DIGITAL AGE (pp. 141–153)

doi: 10.16758/j.cnki.1004-9371.2025.04.016

The niche discipline of Indo-European Studies has proven itself to be prevailingly au courant by launching several projects in the field of online etymological dictionaries. My paper will offer an overview of these projects (including the *Lexicon Etymologicum Digitale Indoeuropaeum* (LEDI) directed by me) and analyse their approaches, features, and peculiarities (e.g., commercial vs. open access). Special attention will be paid to the projects’ inclusions of phonetic rules and affixes, which makes derivation transparent and is helpful for didactic purposes.

Keywords: online dictionary – etymology – open access – phonetic rules – affixes

Péter KATÓ (Eötvös Loránd University Budapest)

GREEK EPIGRAPHIC DATABASES: SCOPE, STRUCTURE, AND PRACTICAL USE (pp. 155–164)

doi: 10.16758/j.cnki.1004-9371.2025.04.017

This paper explores the evolving landscape of digital resources in Greek epigraphy. A selection of digital projects is examined for its structure, accessibility, and scholarly utility, with a critical focus on completeness, editorial policy, and economic sustainability. The analysis reveals tensions between open-access ideals and the realities of commercial publishing, as well as challenges posed by short-term funding and limited project scopes. The paper also considers the integration of artificial intelligence tools, notably Ithaca, assessing their potential and current limitations. Emphasis is placed on the pedagogical impact of digital resources, showing how they empower a new generation of students and democratize access to source material. Ultimately, the study underscores the necessity of balancing innovation with scholarly rigor, advocating for ongoing critical reflection to ensure the meaningful development of digital epigraphy in both research and education.

Keywords: epigraphy – databases – academic research – publishing – democratization

Martina FILOSA (University of Cologne) / **Claudia SODE** (University of Cologne)

THE ROLE OF DIGITAL HUMANITIES IN DIDACTICS: INSIGHTS FROM A SMALL DISCIPLINE IN AN INTERNATIONAL CONTEXT (pp. 165–177)

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This article examines the use of digital tools in research related to and the teaching of Byzantine Studies, a field with a strong scholarly tradition but often limited resources. Drawing on personal experiences from the University of Cologne, the authors present a case study which considers how digital approaches might contribute to the field's sustainability and broader accessibility. This case study focuses on two initiatives – DigiByzSeal and DiBS – which apply digital methods to related areas such as sigillography, epigraphy, numismatics, and manuscript studies, particularly in the context of teaching. Special attention is given to a workshop hosted by the Institute for the History of Ancient Civilizations (IHAC), Northeast Normal University (NENU) in Changchun (China), where students – many of whom being new to the subject – tested XML-based tools in a

classroom setting. The article looks at how this workshop functioned in practice and argues that using digital methods in teaching not only raises the profile of Byzantine Studies but also contributes to broader methodological innovation in the Humanities.

Keywords: Byzantine Studies – sigillography – epigraphy – numismatics – manuscript studies – Digital Humanities – XML technologies – EpiDoc

Stefan KRMNICEK (University of Tübingen)

TEACHING NUMISMATICS THROUGH PLAYFUL APPROACHES

(pp. 179–194)

doi: 10.16758/j.cnki.1004-9371.2025.04.019

This paper presents three computer games that are used in the teaching of ancient numismatics. All three applications – Craveler, NumisGames, and Graveler – were developed within the last years with the aim of teaching students numismatics through a playful approach. The different didactic goals, learning conditions, and outcomes are summarized and discussed.

Keywords: Numismatics – Digital Humanities – Virtual Reality – gaming – teaching – Roman coinage

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EXPRESS XR – EXTENDED REALITY APPLICATIONS FOR MUSEUM EXHIBITIONS (pp. 195–211)

doi: 10.16758/j.cnki.1004-9371.2025.04.020

Extended Reality (XR) technologies are increasingly employed in museums and heritage contexts to create immersive, interactive experiences that enhance public engagement with cultural content. However, their widespread adoption is often hindered by high development costs, limited technical expertise, and accessibility concerns. This paper introduces ExPresS XR (Experimentation and Presentation for Science with OpenXR), an open-source platform built on the Unity game engine, enabling the creation of Virtual, Augmented, and Mixed Reality exhibitions without programming skills. The system provides museum-specific functionalities, including rapid virtual space creation, interactive artifact showcases, gamified learning modules, and integrated visitor data collection. Two case studies – *Temple Tax & Doves*, a VR reconstruction of Herod's Temple

with numismatic learning tasks, and *Hornmoldhaus VR*, a comprehensive virtual representation of the Hornmoldhaus City Museum incorporating both its Japanese artifact collection and other exhibits such as a Roman engraved gem, reconstructed wine cellar, and archival holdings – demonstrate the platform’s application in archaeological and historical contexts. Drawing on over ten XR projects, the paper identifies best practices for motion design, interaction simplicity, and balancing scholarly accuracy with engaging gameplay. ExPresS XR offers a practical, cost-effective approach to XR-based science communication, supporting museums, archaeologists, and cultural institutions in developing accessible, research-informed virtual exhibitions.

Keywords: Extended Reality (XR) – Virtual Reality (VR) exhibitions – Serious Game technologies – digital heritage – object-science communication – digital museums

Sebastian HAGENEUER (Berlin-Brandenburgische Akademie der Wissenschaften)

ARCHAEOGAMING: TEACHING ARCHAEOLOGY THROUGH CREATING VIDEO GAMES (pp. 213–227)

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The present paper explores the innovative use of video game creation, as a pedagogical tool, for teaching archaeology through an interdisciplinary course at the University of Cologne. The course entitled “That belongs in a museum!” is rooted in the concept of “archaeogaming,” as defined by Andrew Reinhard. Students from Archaeology, Digital Humanities and Digital and Computational Archaeology are brought together to collaboratively design video games that convey archaeological knowledge and challenge popular stereotypes. Through practical experience with game engines and storytelling techniques, students go on to create playable games which address themes such as cultural heritage, decolonisation, and public engagement with the past. The main discussion points for this paper will centre on three student projects, which were selected to best illustrate the creative potential and educational value of this approach. This paper contends that the creation of games has the potential to nurture digital literacy and collaborative skills, whilst simultaneously stimulating further critical reflections on archaeological practices and its variable representations in different media. While acknowledging potential challenges in infrastructure and expertise, this study demonstrates that archaeogaming can enrich university curricula by merging academic content with interactive media. This project, therefore, can be regarded as a case study in how digital tools and interdisciplinary cooperation

can enhance both archaeological education and scientific communication(s) in a rapidly evolving media landscape.

Keywords: archaeogaming – Digital Archaeology – game-based learning – interdisciplinary teaching – science communication

Elisabeth GÜNTHER (Institute for Classical Archaeology and Byzantine Archaeology, University of Heidelberg)

TEACHING AI IN ARCHAEOLOGY: PRACTISING PATTERN RECOGNITION WITH ORANGE DATA MINING (pp. 229–246)

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AI applications have already – irreversibly – entered the fields of Humanities, Classics, and archaeological research, but are rarely taught in class. This paper attempts to encourage teachers and students of Classical Archaeology, Classics, Art History, and related fields, to practically explore, evaluate, and critically discuss pattern recognition (here focusing on Greek vase-painting), without requiring previously acquired coding skills. For this task, I will outline the potential of the open access program “Orange Data Mining” for academic teaching, based on a seminar taught by myself at Heidelberg University in the winter semester of 2024/25. I will introduce four different pattern recognition exercises (“Image Grid,” “Image Clustering,” “Image Classification,” and “Prediction”) that are easily accessible for classicists (please, try this at home!) and report the results and experiences which came out of our seminar. Furthermore, I will evaluate how the “hands-on” use of Orange Data Mining in class enables students to access the current debate on the chances and limitations of AI for research in ancient studies.

Keywords: Digital Archaeology – Digital Classics – pattern recognition – AI – teaching